

# Miroslav Šotek

Independent researcher · Founder · Systems engineer

Switzerland · [anulum.li](https://anulum.li) · [GitHub](https://github.com) · [ORCID 0009-0009-3560-0851](https://orcid.org/0009-0009-3560-0851) · [PyPI](https://pypi.org/user/protoscience) · [protoscience@anulum.li](mailto:protoscience@anulum.li)

## ■ PROFILE

Founder and engineering lead of ANULUM CH&LI, an independent research lab. The lab runs on one engine: an autonomous coding and verification system that builds, tests and reviews every project and binds its claims to evidence. The engine is applied in four domains: reactor systems, systems integration and control, neuromorphic computing, and quantum computing. Work runs from mathematical models through reproducible software, native acceleration, formal checks and hardware-facing paths.

## ■ SELECTED WORK

### The engine

- **Synapse Channel:** local-first coordination bus for agent fleets with claims, durable messaging, receipts, audit and federation.
- **Director-AI:** response-level LLM factual-consistency and contradiction guardrail with NLI/RAG evidence.
- **Rigor Foundry:** evidence-bound repository inventory, audit-candidate generation and remediation planning.
- **Remanentia:** auditable memory for AI agents with hybrid retrieval and consolidation.

### Four domains

- **Reactor systems:** SCPN Fusion Core for plasma physics, solvers and validation, and a public catalogue of 21 fusion device families; 29 fission, chemical and hybrid families are planned.
- **Systems integration and control:** SCPN Control, a control-grade runtime with fail-closed admission, and SCPN Phase Orchestrator for synchronisation analysis.
- **Neuromorphic computing:** SC-NeuroCore, a stochastic and neuromorphic stack spanning Python, Rust SIMD, HDC/VSA and RTL generation.
- **Quantum computing:** SCPN Quantum Control, coupled-oscillator experiments on real quantum hardware with protocols committed before the run and hash-bound result packs; IQM granted extended IQM Resonance access for this research.

## ■ COLLABORATION

Open to selective research, engineering, review, deployment and institutional partnership conversations. Organisations can fund defined work under a written research agreement with direct invoicing; AGPL projects are available under commercial licences. Public GitHub currently marks the account as hireable.

## ■ TECHNICAL SCOPE

**Languages:** Python, Rust, TypeScript, JavaScript, Go, Julia, Mojo, C/C++, Lean 4, Verilog, SystemVerilog, WGSL, Shell and LaTeX.

**Systems:** LLM assurance, agent coordination, auditable memory, NLI/RAG, numerical solvers, PyO3/Maturin, SIMD, WebAssembly, WebGPU, FPGA RTL, formal verification, provenance and evidence packaging.

**Quality:** deterministic tests, coverage gates, strict typing where declared, Ruff, Cargo fmt/Clippy, CodeQL, Bandit, SPDX/REUSE, SBOMs, hash-pinned dependencies, protocols committed before the run and replayable result packs.

## ■ PORTFOLIO AND DELIVERY

Maintains a mapped ecosystem of **39 portfolio repositories** across Reactor, Systems Integration and Control, Agentic Assurance, Neuromorphic Computing and Quantum Computing, plus the standalone HushLine utility. The public portfolio contains **34 project entry points** and the PyPI profile contains **19 published projects**, verified 2026-09-25.

## ■ RESEARCH OUTPUT

- 28 ORCID-linked Zenodo records as of 2026-09-25, including technical notes, preprints and software archives. See the [publication index](#).
- SCPN Framework v1.0 technical specification: [10.5281/zenodo.18507639](https://zenodo.org/record/18507639).
- Phase-system matched-false-alarm and grid regime-map preprints: [10.5281/zenodo.22113062](https://zenodo.org/record/22113062) and [10.5281/zenodo.22113116](https://zenodo.org/record/22113116).